

THE LIFTING PROBLEM FOR UNIVERSAL QUADRATIC FORMS IN DEGREES SIX AND SEVEN

ABSTRACT. We show that no totally real number field of degree 6 or 7 admits a universal quadratic form with rational integer coefficients (a universal $\mathbb{Z}$ -form). Together with the classification in degrees ≤ 5 by Kala and Yatsyna, the fields of degree at most 7 with a universal $\mathbb{Z}$ -form are exactly $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$. The proof replaces the house bound of the degree- ≤ 5 argument by the geometry of the *variance lattice* $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$: a tensor-product minimum bound turns the existence of a universal $\mathbb{Z}$ -form into lower bounds for traces of totally positive integers, and a “downspread” argument bounds the shortest vector of Λ_K by $(5 + \sqrt{21})/2$. This makes the search space small: in the case where the shortest vector generates K , no number field has to be computed at all (every candidate dies inside the order $\mathbb{Z}[x_1]$); in the imprimitive case only 18 fields are ever built, and the last five are excluded by explicit elements. A second, independent route through the Mordell–Ko theorem and the E_6 trace threshold confirms every final certificate. In degree 7 (no proper subfields, $\nu(K) \leq (11 + \sqrt{105})/4$) all $5.3 \cdot 10^8$ candidate polynomials are excluded inside $\mathbb{Z}[x_1]$, again without computing a single number field.

1. STATEMENT AND OVERVIEW

Let K be a totally real number field of degree d with ring of integers $\mathcal{O}_K$. A $\mathbb{Z}$ -form is a positive definite quadratic form $Q(x) = \sum_{i \leq j} a_{ij} x_i x_j$ with $a_{ij} \in \mathbb{Z}$; it is *universal over K* if it represents every totally positive element of $\mathcal{O}_K$ over $\mathcal{O}_K$. Kala and Yatsyna [KY2] proved that for $d \leq 5$ such a form exists only for $K = \mathbb{Q}, \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\zeta_7 + \zeta_7^{-1})$. Their proof uses the Mordell–Ko theorem [Ko]: in rank ≤ 5 every positive definite classical integral form is a sum of squares of integral linear forms. This fails in rank 6 (the lattice E_6), which is why degree 6 was the first open case.

Theorem 1.1. *No totally real number field of degree 6 admits a universal $\mathbb{Z}$ -form.*

Theorem 1.2. *No totally real number field of degree 7 admits a universal $\mathbb{Z}$ -form.*

Corollary 1.3. *A totally real field of degree $d \leq 7$ admits a universal $\mathbb{Z}$ -form if and only if it is $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$.*

The proof has a theoretical part (§§2–5), which reduces the theorem to a finite computation with explicit bounds, and a computational part (§6 for degree 6, §7 for degree 7). The main steps are:

- (i) The necessary condition (N) of [KY1]: every totally positive α is $\alpha = \sum M_{ij} \omega_i \omega_j$ for a positive semidefinite half-integral M .
- (ii) A tensor-product bound (Lemma 3.2) turns (N) into a *budget–cost* inequality $\tau(\alpha) \geq \nu(K)$ (Proposition 3.3) and into a *rigidity* statement for elements with $\tau(\alpha) < \frac{3}{2}\nu(K)$ (Proposition 3.4), plus relative versions over subfields.
- (iii) Applying budget–cost to $x_1 - n$ and $-x_1 - n'$, for x_1 a shortest vector of the variance lattice, gives $\nu(K) \leq (5 + \sqrt{21})/2$ (Corollary 4.2). Over a proper subfield, the same argument together with a “positive covering constant” δ_F bounds the relative minimum (Corollary 4.6).
- (iv) The resulting finite enumerations are run in §6. Every candidate field is excluded, and every exclusion is witnessed by an explicit element.

2. SETUP

Throughout, K is totally real of degree d with embeddings $\sigma_1, \dots, \sigma_d$. For $x \in K$ put

$$\tau(x) = \frac{1}{d} \operatorname{Tr}_{K/\mathbb{Q}}(x), \quad \operatorname{Var}(x) = \frac{1}{d} \sum_{j=1}^d (\sigma_j(x) - \tau(x))^2.$$

Both are intrinsic: if x lies in a subfield F , they coincide with the quantities computed in F . The *variance lattice* is $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$ with the positive definite quadratic form $d \operatorname{Var}$; it is a lattice of rank $d-1$ (the embedding $x \mapsto (\sigma_j(x) - \tau(x))_j$ identifies it with a lattice in $\mathbf{1}^\perp \subset \mathbb{R}^d$). Put

$$\nu(K) = \min\{\operatorname{Var}(x) : x \in \mathcal{O}_K \setminus \mathbb{Z}\}, \quad t(K) = \inf\{\tau(\alpha) : \alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}\}.$$

A *shortest vector* is an $x_1 \in \mathcal{O}_K \setminus \mathbb{Z}$ with $\operatorname{Var}(x_1) = \nu(K)$; it is determined by its class up to $x_1 \mapsto \pm x_1 + k$ ($k \in \mathbb{Z}$).

For a subfield $F \subseteq K$ with $f = [F : \mathbb{Q}]$ and $e = [K : F]$, group the embeddings of K into f blocks $B_1, \dots, B_f$ of size e (the fibres over the embeddings $\rho_1, \dots, \rho_f$ of F), and define the block means $\mu_k(x) = \frac{1}{e} \sum_{j \in B_k} \sigma_j(x)$ and

$$\operatorname{Var}_{K/F}(x) = \frac{1}{d} \sum_{k=1}^f \sum_{j \in B_k} (\sigma_j(x) - \mu_k(x))^2, \quad \nu_{K/F} = \min\{\operatorname{Var}_{K/F}(x) : x \in \mathcal{O}_K \setminus F\}.$$

The map $x \mapsto (\sigma_j(x) - \mu_k(x))_{j \in B_k, k}$ has kernel F , and $\mathcal{O}_K/\mathcal{O}_F$ is torsion free, so $\operatorname{Var}_{K/F}$ is a positive definite form on the lattice $\mathcal{O}_K/\mathcal{O}_F$ of rank $d-f$.

3. CONDITION (N) AND THE TENSOR BOUND

Lemma 3.1 ([KY1, Prop. 3.1]). *Let $\omega_1, \dots, \omega_d$ be a $\mathbb{Z}$ -basis of $\mathcal{O}_K$. If K admits a universal $\mathbb{Z}$ -form, then every $\alpha \in \mathcal{O}_K^+$ satisfies*

$$(N) \quad \alpha = \sum_{i,j} M_{ij} \omega_i \omega_j \quad \text{for some positive semidefinite } M = M^t \text{ with } M_{ii} \in \mathbb{Z}, \ 2M_{ij} \in \mathbb{Z}.$$

Proof. If Q has Gram matrix M_Q and $\alpha = Q(v)$ with $v \in \mathcal{O}_K^r$, write $v = V\omega$ with $V \in \mathbb{Z}^{r \times d}$ and take $M = V^t M_Q V$. $\square$

Equivalently, $2\alpha = Q_L(v)$ where $L = (\mathbb{Z}^d, 2M)$ is an *even* lattice. Replacing L by its quotient by the radical, we may take L positive definite of rank $\leq d$; then $\min L \geq 2$, and every vector of L that is not a minimal vector has norm ≥ 4 .

Lemma 3.2 (tensor floor). *Let L, M be positive definite lattices (real quadratic forms on free $\mathbb{Z}$ -modules), and let $w \in L \otimes M$, $w \neq 0$, have tensor rank s . Then $Q(w) \geq \frac{s}{\gamma_s^2} \min L \cdot \min M$, where γ_s is Hermite's constant. In particular, for $s \leq 6$,*

$$Q(w) \geq \min L \cdot \min M, \quad \text{and} \quad Q(w) \geq \frac{3}{2} \min L \cdot \min M \quad \text{if } s \geq 2.$$

Proof. By the Smith normal form, the coefficient matrix of w is $\sum_{i=1}^s x_i y_i^t$ with $x_1, \dots, x_s \in L$ and $y_1, \dots, y_s \in M$ linearly independent. With the Gram matrices X of the x_i and Y of the y_i we have $Q(w) = \operatorname{tr}(XY) = \operatorname{tr}(X^{1/2} Y X^{1/2}) \geq s(\det X \det Y)^{1/s}$ by the arithmetic-geometric mean inequality, and $\det X \geq (\min L / \gamma_s)^s$, $\det Y \geq (\min M / \gamma_s)^s$ by Hermite's inequality for the lattices spanned by the x_i and the y_i . The known values $\gamma_1^2 = 1$, $\gamma_2^2 = \frac{4}{3}$, $\gamma_3^2 = 2^{2/3}$, $\gamma_4^2 = 2$, $\gamma_5^2 = 8^{2/5}$, $\gamma_6^2 = (64/3)^{1/3}$ (see [CS]) give $s/\gamma_s^2 = 1, 1.5, 1.889, 2, 2.176, 2.163$ for $s = 1, \dots, 6$. $\square$

This is the elementary case of Kitaoka's theorem on minima of tensor products [Ki]; we only need tensor rank ≤ 6 , because $\operatorname{rk} \Lambda_K = d-1 \leq 5$ when $d \leq 6$.

Proposition 3.3 (budget-cost). *Let $d \leq 7$ and suppose K admits a universal $\mathbb{Z}$ -form. Then $\tau(\alpha) \geq \nu(K)$ for every $\alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}$; that is, $t(K) \geq \nu(K)$.*

Proof. Write $2\alpha = Q_L(v)$ with $v \in L \otimes \mathcal{O}_K$ as above, and let $m = \frac{1}{d} \sum_j \sigma_j(v) \in L \otimes \mathbb{R}$. Averaging $Q_L(\sigma_j v) = Q_L(m) + 2B(m, \sigma_j v - m) + Q_L(\sigma_j v - m)$ over j ,

$$(1) \quad 2\tau(\alpha) = Q_L(m) + \frac{1}{d} Q_{L \otimes \Lambda_K}(\bar{v}),$$

where $\bar{v}$ is the image of v in $L \otimes \Lambda_K$ and Λ_K carries the form $d \text{Var}$. Since $\alpha \notin \mathbb{Z}$ we have $v \notin L \otimes \mathbb{Z}$, so $\bar{v} \neq 0$ ($\mathbb{Z}$ is saturated in $\mathcal{O}_K$), and Lemma 3.2 gives $Q_{L \otimes \Lambda_K}(\bar{v}) \geq 2 \cdot d\nu(K)$. $\square$

Proposition 3.4 (rigidity). *Under the hypotheses of Proposition 3.3, let $\alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}$ with $\tau(\alpha) < \frac{3}{2}\nu(K)$. Then $\alpha = y^2 + cy + b$ with $y \in \mathcal{O}_K \setminus \mathbb{Z}$, $b, c \in \mathbb{Z}$ and $\nu(K) \leq \text{Var}(y) \leq \tau(\alpha)$.*

Proof. In (1), $\frac{1}{d}Q(\bar{v}) < 3\nu(K)$. By Lemma 3.2, $\bar{v}$ has tensor rank 1, i.e. $\bar{v} = e \otimes \bar{y}$, and $Q_L(e) = 2$ (otherwise $Q_L(e) \geq 4$ and the cost is $\geq 4\nu$). Lifting, $v = e \otimes y + r$ with $r \in L$, so $2\alpha = 2y^2 + 2B(e, r)y + Q_L(r)$ with $Q_L(r) \in 2\mathbb{Z}$. Finally $2\tau(\alpha) \geq \frac{1}{d}Q(\bar{v}) = 2 \text{Var}(y)$. $\square$

Proposition 3.5 (relative versions). *Suppose K ($d \leq 6$) admits a universal $\mathbb{Z}$ -form, and let $F \subsetneq K$ be a subfield. Then $\tau(\alpha) \geq \nu_{K/F}$ for every $\alpha \in \mathcal{O}_K^+ \setminus F$. If moreover $\tau(\alpha) < \frac{3}{2}\nu_{K/F}$, then $\alpha = y^2 + \gamma y + \beta$ with $y \in \mathcal{O}_K \setminus F$, $\beta, \gamma \in \mathcal{O}_F$ and $\text{Var}_{K/F}(y) \leq \tau(\alpha)$.*

Proof. Decompose $\mathbb{R}^d$ orthogonally into block-constant vectors and their complement. The part of $\frac{1}{d}Q(\bar{v})$ in the complement is $\frac{1}{d}Q(\pi\bar{v})$, where $\pi\bar{v} \in L \otimes \pi(\Lambda_K)$ and $\pi(\Lambda_K) \cong \mathcal{O}_K/\mathcal{O}_F$ carries the form $d \text{Var}_{K/F}$ of minimum $d\nu_{K/F}$. If $\alpha \notin F$ then $v \notin L \otimes \mathcal{O}_F$, so $\pi\bar{v} \neq 0$; apply Lemma 3.2 as before. For rigidity, $\pi\bar{v} = e \otimes \pi(\bar{y})$ with $Q_L(e) = 2$. Because $\ker \pi$ on $L \otimes \Lambda_K$ is $L \otimes \Lambda_F$, this gives $v - e \otimes y \in L \otimes \mathcal{O}_F$; expand as before, using that L is even. $\square$

4. BOUNDING THE MINIMA

Lemma 4.1 (downspread). *Let $x \in \mathcal{O}_K \setminus \mathbb{Z}$ with conjugates $\theta_1 < \dots < \theta_d$. Then $\alpha_- = x - (\lceil \theta_1 \rceil - 1)$ and $\alpha_+ = -x + (\lfloor \theta_d \rfloor + 1)$ are totally positive and non-rational, and $\tau(\alpha_-) = \tau(x) - \lceil \theta_1 \rceil + 1 < \tau(x) - \theta_1 + 1$, $\tau(\alpha_+) = \lfloor \theta_d \rfloor + 1 - \tau(x) < \theta_d - \tau(x) + 1$.*

Corollary 4.2. *If K of degree $d \leq 7$ admits a universal $\mathbb{Z}$ -form, then $d\nu \geq 2(\nu - 1)^2$ for $\nu = \nu(K)$. In particular $\nu(K) \leq \frac{(d+4) + \sqrt{(d+4)^2 - 16}}{4}$, which for $d = 6$ is $(5 + \sqrt{21})/2 = 4.7913\dots$*

Proof. Let x_1 be a shortest vector and $w_j = \sigma_j(x_1) - \tau(x_1)$. By Proposition 3.3 and Lemma 4.1, $-\min_j w_j > \nu - 1$ and $\max_j w_j > \nu - 1$. Hence $d\nu = \sum w_j^2 \geq 2(\nu - 1)^2$. $\square$

The enumeration of §6 uses the exact forms of these inequalities (with the integer parts), together with the following consequence of budget-cost for quadratic polynomials in x_1 .

Lemma 4.3. *Let x_1 be a shortest vector of degree ≥ 3 and $P \in \mathbb{Z}[t]$ monic of degree 2 with $P(\sigma_j(x_1)) > 0$ for all j . Under the hypotheses of Proposition 3.3, $P(\tau(x_1)) \geq 0$.*

Proof. $P(x_1)$ is totally positive and not rational, and $\tau(P(x_1)) = \text{Var}(x_1) + P(\tau(x_1))$. $\square$

Definition 4.4. For a totally real field F of degree f with Minkowski lattice $\sigma(\mathcal{O}_F) \subset \mathbb{R}^f$, the positive covering constant is

$$\delta_F = \sup_{u \in \mathbb{R}^f} \min \left\{ \frac{1}{f} \sum_k (p_k - u_k) : p \in \sigma(\mathcal{O}_F), p_k > u_k \text{ for all } k \right\}.$$

For example $\delta_{\mathbb{Q}} = 1$. The function $h(u) = \min_{p > u} \sum_k (p_k - u_k)$ is $\sigma(\mathcal{O}_F)$ -periodic and satisfies $h(u) \leq h(u') + \sum_k (u'_k - u_k)$ whenever $u \leq u'$ coordinatewise. Evaluating h on a grid of step ε that covers a fundamental parallelepiped therefore gives the rigorous bound $\delta_F \leq \max_{\text{grid}} h/f + \varepsilon$ (deltaF.py).

Lemma 4.5 (relative downspread). *For $y \in \mathcal{O}_K \setminus F$ with block minima $m_k = \min_{j \in B_k} \sigma_j(y)$, there is $\gamma \in \mathcal{O}_F$ with $y + \gamma$ totally positive and $\tau(y + \gamma) \leq \frac{1}{f} \sum_k (\mu_k(y) - m_k) + \delta_F$. The same holds for $-y$ with the block maxima.*

Proof. $\tau(y + \gamma) = \frac{1}{f} \sum_k (\mu_k(y) - m_k) + \frac{1}{f} \sum_k (\rho_k(\gamma) + m_k)$, and the second sum can be made $\leq f\delta_F$ with all $\rho_k(\gamma) + m_k > 0$. $\square$

Corollary 4.6. *If K ($d \leq 6$) admits a universal $\mathbb{Z}$ -form and $F \subsetneq K$ with $e = [K : F]$, then $e\nu_{K/F} \geq 2(\nu_{K/F} - \delta_F)^2$.*

Proof. Let y realise $\nu_{K/F}$, and write u_k for the vector of within-block deviations in block k , with maximum M_k and minimum $-m'_k$. By Proposition 3.5 and Lemma 4.5, $D_{\pm} \geq \nu_{K/F} - \delta_F$, where $D_+ = \frac{1}{f} \sum M_k$ and $D_- = \frac{1}{f} \sum m'_k$. Since $\|u_k\|^2 \geq M_k^2 + m_k'^2 \geq (M_k + m'_k)^2/2$, $d\nu_{K/F} = \sum_k \|u_k\|^2 \geq \frac{1}{2f} (\sum_k (M_k + m'_k))^2 \geq 2f(\nu_{K/F} - \delta_F)^2$. $\square$

For $F = \mathbb{Q}$ this is Corollary 4.2 (with $\delta_{\mathbb{Q}} = 1$, $e = d$).

5. RIGIDITY VIA SQUARE ROOTS, AND THE E_6 THRESHOLD

For testing, rigidity is used in the following form, which needs no search in $\mathcal{O}_K$.

Lemma 5.1. *Let α be as in Proposition 3.4 and suppose $\mathbb{Q}(\alpha) = K$. Then there is $\kappa \in \mathbb{Z}$ with $\kappa \equiv 0, 3 \pmod{4}$ such that $4\alpha - \kappa = z^2$ with $z \in \mathcal{O}_K$, $4\nu(K) \leq \text{Var}(z) \leq 4\tau(\alpha)$ and $0 < 4 \min_j \sigma_j(\alpha) - \kappa \leq \max(\frac{d^2}{d-1} \tau(\alpha), \text{Var}(\alpha)/\nu(K))$ (for $d = 6$: $\frac{36}{5} \tau(\alpha) = 7.2 \tau(\alpha)$). Equivalently, the polynomial $4^d m_{\alpha}((t^2 + \kappa)/4) \in \mathbb{Z}[t]$ has an irreducible factor of degree d whose roots have variance in $[4\nu(K), 4\tau(\alpha)]$.*

Proof. Take $z = 2y + c$ and $\kappa = 4b - c^2$. Put $u = 4 \min \sigma(\alpha) - \kappa$, so that $|\sigma_j(z)| \geq \sqrt{u}$. If $p \geq 1$ conjugates of z are positive and $q \geq 1$ negative, then $\text{Var}(z) \geq \frac{pq}{d^2} (2\sqrt{u})^2 \geq \frac{d-1}{d^2} 4u$, so $u \leq \frac{d^2}{d-1} \tau(\alpha)$. If they all agree, $\text{Var}(z) = \text{Var}(\sqrt{4\alpha - \kappa}) \leq 4 \text{Var}(\alpha)/u$, and $\text{Var}(z) \geq 4\nu$ gives $u \leq \text{Var}(\alpha)/\nu$. $\square$

The second tool is used as an independent cross-check of the final certificates. It is the place where E_6 enters.

Proposition 5.2 (E_6 threshold). *Let $d = 6$ and $\alpha \in \mathcal{O}_K^+$ with $\text{Tr } \alpha < 3(3d_K)^{1/6}$. Then (N) holds for α if and only if*

$$(\text{SoS}_2) \quad 2\alpha = \sum_k x_k^2 \quad \text{with } x_k \in \mathcal{O}_K \quad \text{and} \quad \sum_k x_k \in 2\mathcal{O}_K.$$

Proof. If (SoS₂) holds, write $x = U\omega$ and take $M = \frac{1}{2}U^t U$: its diagonal is integral because $\sum_k U_{ki}^2 \equiv \sum_k U_{ki} \pmod{2}$. Conversely, let $2\alpha = Q_L(v)$ with $\tilde{L}$ even of rank $r \leq 6$, and write $v = \sum_k f_k \otimes \omega_k$ with $f_k \in L$. If $f_1, \dots, f_6$ are linearly dependent, then 2α is represented by an even lattice of rank ≤ 5 ; by Mordell–Ko [Ko, Mo] its Gram matrix is $U^t U$ with U integral, which gives (SoS₂). Otherwise $G = (B(f_k, f_l))$ is the Gram matrix of an even positive definite lattice of rank 6. Such a lattice has determinant ≥ 3 : even unimodular positive lattices have rank $\equiv 0 \pmod{8}$ [Se, Ch. V], and an even lattice of rank 6 and determinant 2 has discriminant form $q \equiv \frac{1}{2}$ or $\frac{3}{2} \pmod{2}$; gluing it with E_7 (discriminant form $\frac{3}{2}$), respectively with A_1 (discriminant form $\frac{1}{2}$), would produce an even unimodular lattice of rank 13, respectively 7. Hence $2 \text{Tr } \alpha = \text{tr}(GT) \geq 6(\det G \det T)^{1/6} \geq 6(3d_K)^{1/6}$, where T is the trace form of the basis ($\det T = d_K$). $\square$

The extremal case of the determinant bound is E_6 (determinant 3). It is the only obstruction to the degree- ≤ 5 method in degree 6, but it lives above trace $3(3d_K)^{1/6} \geq 29.47$. Every certificate below has trace ≤ 14 .

6. THE COMPUTATION

All code is in `higher-degree/src.run_degree6.sh` reproduces everything in about 10 minutes on 4 cores; `run_sanity.sh` runs the same machinery in degrees 2–5 (gcc, PARI/GP 2.15, python3 with numpy). Floating point is used only to *locate* candidates, always with margins in the direction that keeps more candidates. Every exclusion is decided in exact arithmetic.

Let K be a sextic field with a universal $\mathbb{Z}$ -form and x_1 a shortest vector. Either $\mathbb{Q}(x_1) = K$ (Case A) or $F = \mathbb{Q}(x_1)$ is a quadratic or cubic subfield (Case B).

6.1. Case A: the shortest vector generates K . Normalise $x_1 \mapsto \pm x_1 + k$ so that $\text{Tr } x_1 \in \{0, 1, 2, 3\}$. By Corollary 4.2, the minimal polynomial $x^6 + a_1x^5 + \cdots + a_6$ of x_1 satisfies $a_1 \in \{0, -1, -2, -3\}$ and $\sum_j (\sigma_j x_1 - \tau x_1)^2 \leq 15 + 3\sqrt{21}$. `enum6.c` enumerates all real-rooted integral polynomials with these constraints. It uses Robinson’s method: by Rolle’s theorem each coefficient is confined to an interval determined by the roots of the previous derivative [Ro, MS]. The program keeps only those whose roots pass the exact downspread inequalities of Lemma 4.1 (with $\nu = \text{Var}(x_1)$) and Lemma 4.3.

real-rooted polynomials enumerated (leaves)	2 088 667
passing downspread and quadratic filters	40 764
irreducible	15 720
NS: a shorter non-rational element exists in $\mathbb{Z}[x_1]$	873
BC: some $\alpha \in \mathbb{Z}[x_1]^+ \setminus \mathbb{Z}$ has $\tau(\alpha) < \text{Var}(x_1)$	5 062
RG: some $\alpha \in \mathbb{Z}[x_1]^+$ in the rigidity window fails Lemma 5.1	9 785
survivors	0

Each of the three order-level tests is a necessary condition for x_1 to be a shortest vector of Λ_K for a field with a universal $\mathbb{Z}$ -form. They are stated in the order $\mathbb{Z}[x_1] \subseteq \mathcal{O}_K$, so no maximal order is ever computed in Case A. The list of the 15 720 polynomials with the reason for each exclusion is `results/prim_status.txt`.

6.2. Case B: the shortest vector lies in a proper subfield. Now $\nu(K) = \nu(F)$, and every element of F is an element of K with the same τ and Var . The same enumeration in degrees 2 and 3 (in degree 2 without Lemma 4.3, which needs $\deg x_1 \geq 3$), using only the NS and BC tests (not rigidity, since the y of Proposition 3.4 may lie outside F), leaves exactly

$$F \in \{\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+\}, \quad \nu(F) = \frac{5}{4}, 2, \frac{14}{9}, 2.$$

For each F let y be a relative shortest vector, so $K = F(y)$ since $[K : F] \in \{2, 3\}$ is prime. Corollary 4.6 and the certified bounds for δ_F give:

F	e	$\delta_F \leq$	$\nu_{K/F} \leq$	candidates	order-level survivors
$\mathbb{Q}(\sqrt{5})$	3	1.3110	3.6513	2504 relative cubics	20
$\mathbb{Q}(\sqrt{2})$	3	1.7091	4.2272	3873 relative cubics	34
$\mathbb{Q}(\zeta_7)^+$	2	1.5513	3.3934	1582 values of Δ	1
$\mathbb{Q}(\zeta_9)^+$	2	2.1529	4.2030	2242 values of Δ	7

Here a relative cubic is $y^3 + b_1y^2 + b_2y + b_3$ over $\mathcal{O}_F$, with b_1 modulo $3\mathcal{O}_F$ and $\frac{1}{6}\text{Tr}_F(\frac{2}{3}b_1^2 - 2b_2) = \text{Var}_{K/F}(y)$. A relative quadratic is $y = (-b_1 + \sqrt{\Delta})/2$ with $\Delta \in \mathcal{O}_F^+$, b_1 modulo $2\mathcal{O}_F$, $\Delta \equiv b_1^2 \pmod{4}$ and $\text{Var}_{K/F}(y) = \text{Tr}_F(\Delta)/12$. The order-level tests in $\mathcal{O}_F[y]$ are the absolute ones (with $\nu(K) = \nu(F)$) together with the relative budget–cost and “ y is relatively shortest”. The 62 survivors give 18 distinct fields. Only these 18 fields are built in the whole proof. On $\mathcal{O}_K$, 12 of them fail relative rigidity (Proposition 3.5) and one fails absolute rigidity. The remaining five fail (N) at an explicit element:

K	d_K	$\text{Tr } \alpha$	$N\alpha$	minimal polynomial of the witness α
$\mathbb{Q}(\sqrt{5}, \zeta_7^+)$	300125	14	29	$x^6 - 14x^5 + 73x^4 - 182x^3 + 227x^2 - 133x + 29$
$\mathbb{Q}(\sqrt{2}, \zeta_7^+)$	1229312	12	8	$x^2 - 4x + 2$ ($\alpha = 2 - \sqrt{2}$)
$\mathbb{Q}(\sqrt{2}, \zeta_9^+)$	3359232	12	8	$x^2 - 4x + 2$ ($\alpha = 2 - \sqrt{2}$)
$x^6 - 9x^4 + 24x^2 - 17$	7138368	12	9	$x^3 - 6x^2 + 9x - 3$ ($\alpha = 2 - \zeta_9 - \zeta_9^{-1}$)
$x^6 - 9x^4 + 24x^2 - 19$	7978176	12	9	$x^3 - 6x^2 + 9x - 3$ ($\alpha = 2 - \zeta_9 - \zeta_9^{-1}$)

Each failure of (N) is established by the exhaustive search of `condN.gp`. It uses $\text{tr } M \leq \text{Tr } \alpha / \lambda_{\min}(T)$, the bound $M_{ii} \leq (\sum_j |\sigma_j(\omega_i^*)| \sqrt{\sigma_j \alpha})^2$ with ω^* the trace-dual basis, integral solution of the linear system for the off-diagonal entries, a Fincke–Pohst enumeration of the solution coset, and an exact positive-semidefiniteness test. The witnesses are checked a second time by Proposition 5.2, with a different search (`sos2.gp`). In the last two fields there is no $x \neq 0$ with $x^2 \preceq 2\alpha$ at all, and in $\mathbb{Q}(\sqrt{5}, \zeta_7^+)$ the only such x are ± 1 . The first witness coincides with the one found earlier by independent code in the working pack [Pack].

6.3. Proof of Theorem 1.1. Suppose K is sextic with a universal $\mathbb{Z}$ -form and let x_1 be a shortest vector. In Case A, the minimal polynomial of a suitable normalisation of x_1 is among the 15 720 polynomials of §6.1, and each of them violates a condition that a shortest vector must satisfy. In Case B, $F = \mathbb{Q}(x_1)$ is one of the four fields listed, and $K = F(y)$ for a relative shortest vector y within the enumerated range. The resulting field is one of the 18 fields of §6.2, each of which violates rigidity or (N). $\square$

7. DEGREE SEVEN

Let K be a septic field with a universal $\mathbb{Z}$ -form. Budget–cost, rigidity and Corollary 4.2 hold for $d = 7$: the tensor rank is at most $\text{rk } \Lambda_K = 6$, and $s/\gamma_s^2 \geq \frac{3}{2}$ for $2 \leq s \leq 6$. Thus $\nu(K) \leq (11 + \sqrt{105})/4 = 5.3117\dots$ Since 7 is prime, K has no proper subfields. So the shortest vector x_1 generates K (only Case A occurs), and every non-rational element of K generates K . In particular Lemma 5.1 applies to every totally positive non-rational α in the rigidity window.

The enumeration of §6.1 (with $\text{Tr } x_1 \in \{0, 1, 2, 3\}$ and centred $T_2 \leq 7(11 + \sqrt{105})/4$) produces $5.3 \cdot 10^8$ leaves. So we also run two of the order-level tests inside the enumerator (`enum6.c`, mode `strong`):

- (NS) for $y = cx_1^2 + bx_1$ ($1 \leq c \leq 3$) and $y = x_1^3 + ax_1^2 + bx_1$, with the integers a, b near the real minimiser of $\text{Var}(y)$: if $0 < \text{Var}(y) < \text{Var}(x_1)$, then x_1 is not a shortest vector;
- (RIG) Lemma 5.1 for the two downspread elements $\alpha_{\pm}$ of Lemma 4.1, whenever $\tau(\alpha_{\pm}) < \frac{3}{2} \text{Var}(x_1)$. As $K = \mathbb{Q}(\alpha)$, the conjugates of z are $\pm \sqrt{4\sigma_j(\alpha) - \kappa}$. So $z \in \mathcal{O}_K$ exists iff, for some κ in the range of the lemma and some choice of signs, all elementary symmetric functions of these numbers are integers, and $\text{Var}(z) \in [4 \text{Var}(x_1), 4\tau(\alpha)]$.

real-rooted polynomials enumerated (leaves)	533 562 406
excluded by the downspread inequalities (Lemma 4.1)	531 897 850
excluded by Lemma 4.3	41 498
excluded by (NS) in the enumerator	2 532
excluded by (RIG) in the enumerator	1 619 004
remaining	1 522
irreducible among them	23
exact tests in $\mathbb{Z}[x_1]$ (§6.1): NS / BC / RG	5 / 3 / 15
survivors	0

Proof of Theorem 1.2. A suitable normalisation of the shortest vector x_1 of K has its minimal polynomial among the enumerated leaves. Every irreducible leaf violates a necessary condition for being the minimal polynomial of a shortest vector of a field with a universal $\mathbb{Z}$ -form. $\square$

No number field and no maximal order is computed in degree 7. The in-enumerator tests (NS), (RIG) are the only new ingredients. Each exclusion they make on an irreducible polynomial was re-checked in exact arithmetic (`verify_strong.gp`): for (NS), the exact minimum of Var on $\mathbb{Z}[x_1] \setminus \mathbb{Z}$ by Fincke–Pohst; for (RIG), `sqshape`, i.e. factorisation of $4^7 m_\alpha((t^2 + \kappa)/4)$ over $\mathbb{Q}$. In degree 7 this re-check covers 857 103 irreducible polynomials (the other 764 433 excluded polynomials are reducible), with no disagreement. In degree 6 it covers all 15 714 exclusions on irreducible polynomials made by the strong mode, also with no disagreement. `run_degree7.sh` reproduces the computation in about 30 minutes on 4 cores.

8. SHARPENINGS

This section records improvements found after the proofs above. They are not needed for Theorems 1.1 and 1.2. They shrink the computation considerably, and they are the starting point for degree 8.

8.1. A single universal form. Oh [Oh] determined minimal-rank n -universal lattices: $E_6 \perp I_7$ and $E_7 \perp I_6$ are 6-universal [Oh, Thms. 2.3, 2.5]; $E_8 \perp I_7$ and $E_7 \perp I_8$ are 7-universal [Oh, Cor. 3.2, Thm. 3.3]; $E_8 \perp I_8$ is 8-universal [Oh, Thm. 3.1]; and I_{n+3} is n -universal for $n \leq 5$.

Proposition 8.1. *Let $d \leq 8$, and let U_d be I_8 ($d \leq 5$), $E_6 \perp I_7$ ($d = 6$), $E_8 \perp I_7$ ($d = 7$), or $E_8 \perp I_8$ ($d = 8$). Let U_d^{ev} be its even sublattice: D_8 , $E_6 \perp D_7$, $E_8 \perp D_7$, $E_8 \perp D_8$ respectively. For $\alpha \in \mathcal{O}_K^+$, condition (N) holds if and only if α is represented over $\mathcal{O}_K$ by the $\mathbb{Z}$ -form $\Phi_d = \frac{1}{2}Q_{U_d^{\text{ev}}}$. In particular, K admits a universal $\mathbb{Z}$ -form if and only if Φ_d is universal over K .*

Proof. If (N) holds, then $2\alpha = Q_L(v)$ with L even, positive definite, of rank $\leq d$. By d -universality there is an isometric embedding $L \rightarrow U_d$, and its image consists of vectors of even norm, hence lies in U_d^{ev} . The Gram matrix of Φ_d is half-integral with integral diagonal, so the converse is Lemma 3.1 applied to $Q = \Phi_d$. $\square$

So in degree 6, besides the sums of squares with parity (the D_7 part) only one extra lattice appears, and it is E_6 . This is the structural form of Proposition 5.2. (N) can also be decided by searching representations by one fixed form of rank ≤ 16 .

8.2. Sharpened rigidity and the square dichotomy. In the proof of Proposition 3.4, $c = B(e, r)$ and $b = Q_L(r)/2$, so by the Cauchy–Schwarz inequality in L

$$\kappa := 4b - c^2 = Q_L(e)Q_L(r) - B(e, r)^2 \geq 0.$$

Moreover $\tau(\alpha) = \text{Var}(y) + (\tau(y) + c/2)^2 + \kappa/4$, where $\text{Var}(y) \geq \nu(K)$.

Proposition 8.2. *Under the hypotheses of Proposition 3.4 we have $4\alpha = z^2 + \kappa$ with $z \in \mathcal{O}_K$, $\kappa \in \mathbb{Z}_{\geq 0}$, $\kappa \equiv 0, 3 \pmod{4}$ and $\kappa + \tau(z)^2 \leq 4(\tau(\alpha) - \nu(K))$. Consequently:*

- (a) (square dichotomy) *if $\tau(\alpha) < \min(\frac{3}{2}\nu(K), \nu(K) + \frac{3}{4})$, then α is a square in $\mathcal{O}_K$;*
- (b) (relative version) *if $F \subsetneq K$ and $\alpha \notin F$ has $\tau(\alpha) < \frac{3}{2}\nu_{K/F}$, then $\alpha = y^2 + \gamma y + \beta$ as in Proposition 3.5, with $4\beta - \gamma^2$ totally non-negative in $\mathcal{O}_F$.*

Proof. If $\kappa \neq 0$ then $\kappa \geq 3$ and $\tau(\alpha) \geq \nu + \frac{3}{4}$. If $\kappa = 0$, then c is even and $\alpha = (y + c/2)^2$. For (b), apply Cauchy–Schwarz at every real place of F . $\square$

The effect is visible at once. In degrees 2 and 3, the order-level filters with the sharpened test leave exactly $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7)^+$, so no (N) search is needed: $2 \pm \sqrt{2}$ and $2 - \zeta_9 - \zeta_9^{-1}$ are window elements that are not squares. In the degree-6 Case B, 17 of the 18 fields fail the sharpened rigidity (`sharp.gp`), and only $\mathbb{Q}(\sqrt{5}, \zeta_7^+)$ needs condition (N).

The square dichotomy also constrains Case B before any enumeration. If $x_1 \in F$, then $\nu(K) = \nu(F)$ and every window element of F must be a square in K . If $[K : F]$ is odd, it must already be a square in F . If $[K : F] = 2$, a non-square α forces $K = F(\sqrt{\alpha})$. In degree 6 this gives the following. $F = \mathbb{Q}(\sqrt{2})$ is impossible, since it would force $\mathbb{Q}(\zeta_{16})^+ \subset K$. $F = \mathbb{Q}(\zeta_9)^+$ forces $K = F(\sqrt{2 - \zeta_9 - \zeta_9^{-1}}) = \mathbb{Q}(\zeta_{36})^+$, which fails the sharpened rigidity. Only $F = \mathbb{Q}(\sqrt{5})$ and $F = \mathbb{Q}(\zeta_7)^+$ need a relative enumeration.

8.3. Integrality-refined bound for ν . With the integer parts kept, the downspread conditions read $\theta_1 \leq L = \lfloor \tau - \nu + 1 \rfloor$ and $\theta_d \geq U = \lceil \tau + \nu - 1 \rceil$. With $a = \tau - L$ and $b = U - \tau$, Cauchy–Schwarz on the middle conjugates gives $d\nu \geq a^2 + b^2 + (a - b)^2/(d - 2)$. For each trace class this is a smaller range (`refined_bound.py`):

d	Corollary 4.2	refined, $\text{Tr } x_1 = 0, 1, 2, \dots$
6	4.791	4.000, 4.139, 4.222, 4.250
7	5.312	4.857, 4.694, 4.204, 4.245
8	5.828	5.000, 5.109, 5.188, 5.234, 5.500

8.4. Pruning the enumeration. The same conditions give $\theta_1 - \tau \leq -a$ and $\theta_d - \tau \geq b$. Two consequences can be tested inside the Rolle tree, and they remove only subtrees without admissible leaves:

- every even centred power sum satisfies $M_k = \sum_j (\theta_j - \tau)^k \geq a^k + b^k$, and M_4, M_6 are fixed at depths 4, 6;
- the middle conjugates lie in $\tau \pm r$ with $r^2 = S - a^2 - b^2$. Since the i -th root of $f^{(d-m)}$ lies in $[\theta_i, \theta_{i+d-m}]$, the roots $2, \dots, m - 1$ of g_m lie in that window.

The enumerator output is unchanged bit for bit. In degree 6 the number of leaves drops from 2 088 667 to 178 438. In degree 7 it drops from 533 562 406 to 7 577 940, and the running time from about 25 minutes to about 1 minute on 4 cores.

8.5. Degree 8: what remains. Everything above holds for $d = 8$: tensor rank ≤ 7 gives $s/\gamma_s^2 \geq \frac{3}{2}$.

- *Case B, quartic F .* The enumeration in degree 4 with the NS and BC tests leaves 10 quartic fields. For 7 of them the window non-squares cannot all become squares in one quadratic extension, so they are impossible. $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ is impossible because its only shortest vector lies in $\mathbb{Q}(\sqrt{5})$. $\mathbb{Q}(\zeta_{16})^+$ forces $K = \mathbb{Q}(\zeta_{32})^+$ and $\mathbb{Q}(\zeta_{24})^+$ forces $K = \mathbb{Q}(\zeta_{48})^+$; both fail the sharpened rigidity.
- *Case B, quadratic F .* $F = \mathbb{Q}(\sqrt{2})$ forces $\mathbb{Q}(\zeta_{16})^+ \subset K$, and hence $K = \mathbb{Q}(\zeta_{32})^+$. $F = \mathbb{Q}(\sqrt{5})$ remains: it needs an enumeration of relative quartics with $\nu_{K/F} \leq 4.21$ (Corollary 4.6, $e = 4$).
- *Case A.* Octic polynomials with $\nu \leq 5.5$ (refined bound; for $\text{Tr } x_1 = 0$ this means $a_2 \geq -20$). With the pruning and the sharpened tests, the subtree $(a_1, a_2) = (0, -12)$ takes 40 seconds and $(0, -14)$ takes 143 seconds ($1.15 \cdot 10^7$ leaves), but $(0, -16)$ did not finish within 25 minutes. Extrapolating, the full run is days to weeks on 4 cores. In the middle range of ν the downspread conditions are not yet restrictive enough to prune the tree, so degree 8 needs a further theoretical input before Case A is practical.

9. CONSISTENCY CHECKS AND STATUS

- **Enumerator.** `enum6.c` agrees exactly with an exact-arithmetic re-implementation (`enum_check.gp`, real-rootedness by Sturm sequences) on all 56 subtrees and all 2 088 667 leaves in degree 6, and

with brute force over coefficient boxes in degrees 3–5. In degree 7 it agrees with `enum.check.gp` on the 48 subtrees with the fewest leaves (2 267 434 leaves in total).

- **Degrees ≤ 5 reproduced.** The same pipeline, with the degree- d bound of Corollary 4.2, leaves $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ in degree 2, and $\mathbb{Q}(\zeta_9)^+$, $\mathbb{Q}(\zeta_7)^+$ in degree 3. Of these, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\zeta_9)^+$ fail (N) at $2 - \sqrt{2}$ and $2 - \zeta_9 - \zeta_9^{-1}$. Nothing is left in degree 5. In degree 4, Case A leaves nothing, and Case B leaves three fields ($d_K = 725, 1600, 7168$), all failing (N). This recovers [KY2, Thm. 1.2].
- **Controls.** (N) holds for all tested elements of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7)^+$; both fields survive every filter.
- **Floating point.** Used for embeddings (60 digits), for root brackets in the enumerator (80-bit, with margins), and for δ_F (double precision with a grid margin of 0.002 resp. 0.02). The grid values $\delta_{\mathbb{Q}(\sqrt{5})} \approx 1.3090 = \varphi^2/2$ and $\delta_{\mathbb{Q}(\sqrt{2})} \approx 1.7071 = 1 + 1/\sqrt{2}$ are consistent with exact values.
- **Recommended before publication.** An independent re-implementation of Case A and of the (N) searches (for example in Magma), and an exact-arithmetic computation of δ_F for the four fields.

Remark 9.1 (degree 8 and beyond). Lemma 3.2 gives $s/\gamma_s^2 \geq \frac{3}{2}$ also for $s = 7, 8$ ($\gamma_7^7 = 64$, $\gamma_8 = 2$). So budget–cost, rigidity and Corollary 4.2 hold for $d \leq 9$, and $d = 8$ has $\nu(K) \leq 3 + 2\sqrt{2} = 5.83$. Two things are new in degree 8. Case B must handle quadratic subfields with relative degree 4 and quartic subfields with relative degree 2. And the Case A enumeration is much larger: the leaf count grows by roughly 2.5–3 per unit of a_2 , and the range of a_2 grows with the bound on ν . The in-enumerator tests of §7 are what keeps the output manageable.

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